

Aditya Kulkarni

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PROFESSIONAL SUMMARY

MS CS candidate (NC State, May 2026) with full-stack product engineering experience across **Python, TypeScript, and React**. Shipped 10+ features across 7 production deployments at Isomer AI (394 commits); built a 3-vendor LLM quality estimation harness tracking hallucination rate, p50/p95 latency, and cost/call on a 2,000-example benchmark; and designed a per-tenant RAG system for compliance-sensitive environments. Experience processing data at scale: 500GB cloud genomics platform on GCP and Kafka + Spark pipelines reducing latency 20% at Capgemini. Track record of completing full-stack MVPs in 48-hour builds.

CORE COMPETENCIES

Python & TypeScript

React & Full-Stack

ML Tooling & Quality Estimation

RAG & LLM Applications

Data Processing at Scale

Production Software Delivery

Cloud (AWS / GCP)

Fast Prototyping

WORK EXPERIENCE

Isomer AI

May 2025 - Present

Product Engineer Intern

- Shipped 10+ production features in **TypeScript, Node.js, Express, PostgreSQL, and React** for an AI-powered claims automation platform -- **394 commits, 7 production deployments**; owned features end-to-end from database schema to UI
- Built multi-model AI agent tooling across **3 LLM providers** (Anthropic Claude, OpenAI GPT-4o, Google Gemini) with MCP, Temporal.io workflow orchestration, real-time event pipelines, RBAC, and document processing (PDF, email, S3); integrated Sentry for production observability

NC State University

Apr 2025 - Present

Graduate Research Assistant

- Designed and deployed a **500GB cloud genomics platform** on Google Cloud Run serving 100+ researchers worldwide; applied hyperparameter tuning and time-series analysis to improve model accuracy 12%

Mphasis (HP Company)

Jun 2024 - Aug 2024

Software Developer Intern

- Built 5+ AI-driven workflows including an OCR + NLP health claim automation tool improving claim accuracy **25%**; automated deployments on AWS and GCP; integrated Grafana monitoring dashboards

Capgemini

Jun 2023 - Aug 2023

AI/ML Intern

- Built distributed data pipelines using **Hadoop + Spark** for batch processing and **Kafka** for real-time ingestion -- reduced ticket-processing latency **20%** and end-to-end processing time 15%; led full SDLC from design through production

PROJECTS

LLM Eval Harness -- Quality Estimation Benchmark Independent, 2025

Built a vendor-neutral quality estimation harness comparing Claude Sonnet 4.6, GPT-4o, and Gemini 2.5 Pro on a **2,000-example frozen test set** (grounded QA, multi-hop QA, FEVER claim verification). Tracked ExactMatch, TokenF1, abstention accuracy, grounded hallucination rate, p50/p95 latency, and cost/call; statistical significance via paired bootstrap CIs and McNemar test. DuckDB + Parquet analytics, Streamlit dashboard for visualization.

Python · Anthropic/OpenAI/Google SDKs · DuckDB · Parquet · Streamlit · Docker

AlgoPulse -- Full-Stack Event-Driven Platform Independent, 2025

Identified API-polling bottleneck under load and re-architected to event-driven streaming. **TypeScript + React** frontend with Supabase Edge Functions; Temporal.io durable workflow state machines and Kafka for backend message brokering -- sustained **2,000 async requests/second** with zero message loss under failure injection.

TypeScript · React · Node.js · Temporal.io · Kafka · Supabase Edge Functions

Glance AI -- Privacy-First RAG System Independent, 2025

Designed a **per-tenant data isolation** architecture at the vector retrieval layer for fintech/healthcare environments -- zero cross-tenant context leakage by design. Deterministic output architecture for factual correctness in regulated data environments.

Python · Vector Databases · RAG · LangChain · LLMs

PerCent -- GenAI Finance App HackNC24 Top 10 of 100+ teams, 48h build

Shipped a complete full-stack GenAI finance application in **48 hours**: goal tracking, expense logging, and a RAG-based AI financial advisor using Qwen-14B-Chat via Hugging Face -- reduced manual financial research **30%**. Ranked Top 10 of 100+ teams.

Python · React · RAG · Hugging Face · Selenium · REST APIs

EDUCATION

MS Computer Science NC State University

Aug 2024 - May 2026

Operating Systems, Algorithms, Software Engineering, Data Science, Distributed Systems, Artificial Intelligence

B.Tech Computer Science Mumbai University

Aug 2020 - May 2024

CGPA 9.32/10 · Big Data Infrastructure, DS & Algorithms, Machine Learning, Software Engineering, Blockchain

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C, Java, R

Frontend & Backend: React, Node.js, Express, REST APIs, PostgreSQL, Redis, MongoDB (adjacent), Prisma, Docker

Data & ML: Kafka, Spark, Hadoop, DuckDB, Parquet, RAG, LangChain, Temporal.io, Streamlit

Cloud & Observability: AWS, GCP, Kubernetes, CI/CD, Grafana, Sentry, Cloud Run, Terraform

Awards: Top 10 in 6+ national hackathons · Google Developer Student Club Lead (500+ students) · IEEE Published